

Spencer McGrath

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OBJECTIVE

Attend WPI, majoring in Robotics Engineering and minoring in Education to earn a Master's in Learning and Education.

EXPERIENCE

Intern *Clarity Cyber* 2022-06 2024-01

- Developed Android applications for government devices
- Created server-side applications supporting DARPA contracts

Participant *NASA Hunch Program* 2023-09 2024-07

- Developed AI assistant for astronauts in deep space to mitigate the effects of long-term isolation
- Extensive ethical research to provide best possible care outcomes
- Presented to industry professionals at Rocket Park in Houston, TX

Mentor/Captain *FRC-888 Robotiators, Glenelg High School* 2020-08 2024-08

- Designed and implemented Java code for dynamic mechanical systems, PID control loops, visual feedback/feedforward control, and trajectory planning with omnidirectional drivebase
- Electrical design for 2021 - 2024 seasons
- Presented to Maryland Congressional Legislatures at the National Advocacy Conference in 2022
- Secured over \$40,000 in grant funding for the team
- Captain for 2023-2024 seasons

EDUCATION

Worcester Polytechnic Institute

Robotics Engineering (Major) 2028-05 3.30

Worcester Polytechnic Institute

Teacher Prep Program (Minor) 2028-05

VOLUNTEERING

- *National Technical Honor Society*

Provided tutoring for fellow high school students for any PLTW or Computer Science classes

- *Ript Events*

Assisted with triathlon and multisport events logistics. As well as lead audio technician for 3 years

- *Glenelg Boosters Club*

Staffed concession and fry stands at local Maryland events

- *Work On Worcester 2024 and 2025*

Assisted local elementary school with repainting in 2024. Helped Pawsitively4Pink in creating crafts for fundraising events.

SKILLS

Technical

Object-Oriented: Java (Expert), C++ (Intermediate)

Hybrid: Python (Intermediate)

Git (Advanced)

CAD (SolidWorks, Fusion360, Onshape) (Intermediate)

AWARDS

- NASA HUNCH National Finalist
- FIRST Robotics Dean's List Semi-Finalist 2023

REFERENCES

Will provide references upon request